TEXORA · LECTURE SHEET

Recipe Prediction for Denim Washing

A denim wash recipe is predicted from a washed-down standard: the wash master works backwards from the target shade and hand-feel to enzyme, stone, bleach and time. The arithmetic is garment arithmetic — loads, ratios and available chlorine.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 5 · SPECIALIZATION — WET PROCESSING

1. The wash as a sequence of predicted steps

Denim washing removes indigo from the surface of the warp yarns to produce a faded, softened garment. A full recipe is a chain of operations, each with its own dosing logic, and the prediction is made from a **wash standard** — an approved garment or leg panel — exactly as dyeing prediction starts from a swatch (Topic 1). The usual chain:

1. **Desizing:** α-amylase removes the starch size so later steps act on indigo, not on size film.
2. **Abrasion:** pumice stone and/or cellulase enzyme abrades the yarn surface and releases indigo (stone wash, enzyme wash, stone–enzyme wash).
3. **Bleach (optional):** sodium hypochlorite or potassium permanganate lightens beyond what abrasion gives (bleach wash, PP spray for local effects).
4. **Neutralisation and clearing:** destroys residual bleach, removes loose indigo and prevents back-staining.
5. **Softening and drying:** cationic softener for hand; tumble dry sets the final appearance.

2. Dosing bases in the wash house

- **Enzymes** are dosed % owf (on weight of garments) — their action is on the goods.
- **Pumice stone** is dosed as a mass ratio to garments (typically 0.5:1 to 1:1 for a medium stone wash; heavier ratios give faster, harsher abrasion and more machine wear).
- **Liquid chemicals** (anti-back-staining agent, neutraliser) are dosed g/L of bath.
- **Hypochlorite bleach** is dosed by **available chlorine** in the bath (g/L as Cl), computed from the commercial bleach's stated available-chlorine content — never by product volume alone, because bleach liquor loses strength in storage.

Bath (L) = garment load (kg) × LR | Available Cl needed (kg) = target (g/L) × bath (L) / 1000 | Bleach product (kg) = Cl needed / available-Cl fraction

3. Fully worked recipe — stone–enzyme wash with bleach

WORKED EXAMPLE 1 — 120 KG OF JEANS, WASHER AT LR 1:10

Bath = 120 × 10 = **1,200 L**.

Desize: α-amylase 1.5 g/L → 1.5 × 1,200 = **1.8 kg**; 60 °C, 20 min, drain.

Stone–enzyme abrasion: cellulase 1.2% owf → 1.2 × 120/100 = **1.44 kg**; pumice at 0.5 kg per kg of garments → **60 kg**; anti-back-staining agent 1 g/L → **1.2 kg**; 50 °C, 40 min (typical values for a medium wash; exact temperature/time follow the enzyme supplier's window and the standard's degree of abrasion).

Bleach: target available chlorine 2.5 g/L → Cl required = 2.5 × 1,200/1000 = **3.0 kg**. Commercial bleach at 12% available chlorine → product = 3.0/0.12 = **25 kg**. Check the bleach's actual strength on receipt — at 10% the same bath would need 30 kg, and dosing 25 kg would under-bleach the load.

Neutralise: sodium metabisulphite 2 g/L → **2.4 kg**, followed by clearing rinse, softener, hydro-extraction and tumble drying.

Every figure is per load: the wash house schedules loads, so recipes are predicted, issued and costed per load, not per kilogram.

4. Local effects — PP spray prediction

Potassium permanganate spray creates the bright whisker and hand-sand areas: a solution of about 1.5% $KMnO_4$ (15 g/L) is sprayed on the inflated garment, dried, then neutralised in the first wash bath (metabisulphite or oxalic acid) so the brown MnO_2 stain converts to a bleached patch. Prediction here is by calibration panel: spray concentration, pressure and distance are fixed by spraying a graded series on test legs and matching the standard — the same build-up discipline as Topic 1, applied to a spray gun. Laser and ozone systems replace this chemistry for many designs (Topic 11), with intensity settings calibrated the same way.

5. Controlling the prediction in production

VARIABLE	WHY PREDICTION FAILS WITHOUT IT
Garment load and LR	Underloading raises abrasion per garment and lightens the wash; the recipe assumes the stated load.

Indigo lot of the denim	Dyeing depth and indigo ring-dyeing quality vary between fabric lots; a recipe predicted on one lot over- or under-washes another. Wash a first-load approval and adjust time/enzyme before running bulk.
Stone condition	Pumice wears and breaks down; recycled stone abrades less per kilogram. Stone ratio is by effective abrasion, not just mass.
Enzyme activity	Enzymes are activity-graded products; a new lot at the same % owf can act stronger. Dose by the supplier's activity basis and re-prove on lot change.
Back-staining	Released indigo re-deposits on weft and pocketing, dulling contrast. Anti-back-staining agent, bath changes and prompt clearing protect the prediction's contrast.

6. Glossary — key terms

- **Available chlorine:** the oxidising power of a bleach liquor expressed as equivalent chlorine, g/L or %.
- **Back-staining:** re-deposition of removed indigo onto lighter parts of the garment.
- **PP spray:** potassium permanganate sprayed locally to create bleached effects after neutralisation.
- **Stone wash:** abrasion of garments with pumice to release surface indigo.
- **Leg panel:** a single garment leg washed as a quick approval sample.

7. Key points

- Denim recipes are predicted per load: enzymes % owf, stone by mass ratio, bath chemicals g/L, bleach by available chlorine.
- Bleach product quantity = required chlorine ÷ actual available-chlorine fraction — verify the bleach strength, do not trust the drum label of an old stock.
- First-load approval against the standard is part of the recipe; lot changes of denim, stone or enzyme reopen the prediction.

8. Summary

Denim washing looks empirical but runs on the same prediction discipline as dyeing: a standard, a calibrated series, a computed recipe (1.44 kg cellulase, 60 kg pumice, 25 kg of 12% bleach for the worked 120 kg load) and controlled variables. The wash master's craft is knowing which variable moved when production disagrees with the prediction.

9. Practice

1. A 100 kg load is washed at LR 1:12. Cellulase is dosed 1.0% owf and pumice at 0.75 kg/kg. Calculate bath volume, enzyme and pumice quantities.
2. A bleach bath of 900 L must hold 2.0 g/L available chlorine. The bleach in store assays at 11% available chlorine (not the 13% on its old label). How much product is needed, and what error would the label value have caused?
3. Why is hypochlorite specified by available chlorine rather than by litres of product per load?
4. A bulk lot washes darker than the standard at the frozen recipe. Give three probable causes and the first correction for each.

Answer key

1. Bath = 100 × 12 = 1,200 L. Enzyme = 1.0% × 100 kg = 1.0 kg. Pumice = 0.75 × 100 = 75 kg.
2. Cl required = 2.0 × 900/1000 = 1.8 kg → product = 1.8/0.11 = 16.4 kg. Using 13% would have dosed 1.8/0.13 = 13.8 kg, delivering only 13.8 × 0.11 = 1.52 kg Cl (1.69 g/L) — an under-bleached load.
3. Hypochlorite liquor decomposes in storage, so equal volumes of product do not carry equal oxidising power. Available chlorine states the active content, making the dose independent of the product's age and brand.
4. Any three: denim lot deeper or slower to abrade (extend abrasion time / verify with leg panel); enzyme lot less active (re-prove dose on activity basis); stone worn or load heavier than recipe (restore stone ratio / correct load); bleach strength decayed (assay and correct by available chlorine).
